MODDIV-BLINDAGE C1: Module Rounding, Dependent-Remainder Correctness, and Conditional Classical-ROM CCA Security

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Scope notice. This article studies only the frozen C1-P12-CBD1 profile. Its correctness certificate is exact, but its confidentiality conclusions are conditional on two explicitly defined Module-LWE source problems and, for the KEM, the classical programmable random-oracle model. No concrete 128-bit certificate, QROM theorem, production-side-channel claim, or security transfer to the later CBD2 profile is asserted.

Abstract

We study the C1-P12-CBD1 profile of MODDIV-BLINDAGE, a module encryption construction over the negacyclic ring $R = \mathbb{Z}_{4096}[x]/(x^{256}+1)$ with rank three, explicit centered-binomial errors, coefficientwise modular division, randomized quotient lifting, and an implicit-rejection KEM wrapper. The correctness proof retains the actual dependent rounding remainders instead of replacing them by fictitious independent noise. An ordered exponential-moment argument gives an exact rational certificate for the averaged worst-message failure probability, $\delta < 2^{-282}$, while a legal event of probability 2^{-6400} proves that perfect correctness is false. Confidentiality is reduced by two distribution-preserving transitions to two precisely specified small-secret Module-LWE source problems with three and four rows. Seeded matrix generation is handled by programmable-oracle simulation of the full raw response, including discarded high bits. Applying the corrected modular Fujisaki–Okamoto analysis yields a conditional single-user IND-CCA bound in the classical random-oracle model with explicit correctness and query terms. The public-key and ciphertext payloads are 993 and 1408 bytes, respectively. These sizes and exact certificates do not establish concrete security; the two source-hardness assumptions remain external obligations.

Relationship to prior work. Module-LWE encryption, coefficient compression and Fujisaki–Okamoto transformations are established techniques, as illustrated by Kyber and related module-lattice KEMs [2]. Power-of-two module rounding and CCA-secure KEM composition are also prior in Saber [3], while randomized preimage lifting appears in middle-product rounding encryption [4]. The contribution here is not any of those ingredients in isolation, but the exact C1 combination: its dependent-remainder correctness certificate, two source simulations, oracle-programmed seeded representation, and the resulting conditional bound. Recent work on concrete MLWE/LWE hardness gaps [5] reinforces the need to preserve the exact module structure in any concrete analysis; it does not provide a security estimate for C1. The MODDIV naming lineage descends from earlier public work [6]; no security theorem from that historical construction is transferred here.

1. What the BLINDAGE claim consists of

Four distinct questions must remain visible: whether the algorithms recover an honest message; whether an adversary is transformed into a solver for an exact source problem; whether that

source is hard at the stated parameters; and whether a concrete byte implementation realizes the proof's model. A small failure bound answers the first question, not the other three.

The C1 profile has degree 256, module rank 3, coefficient modulus 4096, public-key divisor 4, ciphertext divisor 2, and centered-binomial parameter one for every secret and explicit error. Its seeded public-key payload has 993 bytes and its ciphertext 1408 bytes. These are format facts, not bit-security levels. The description uses an ideal matrix-expansion oracle when discussing seeded public-key security. Abstract full-matrix encryption requires no such oracle.

Module encryption, coefficient compression, short errors and FO transforms have established precedents, including Kyber and Saber [2,3]. Randomized preimage lifting also appears in middle-product rounding encryption [4]. The present results concern the exact combination and distributions specified below. They do not establish a new invention of these ingredients, and no exhaustive priority search is claimed.

2. Negacyclic algebra and the public matrix

Set

$$R = \mathbb{Z}_P[x]/(x^n + 1), \qquad n = 256,\ k = 3,\ P = 4096,\ Q = 4,\ C = 2. \tag{1}$$

Every polynomial is a fixed array of n coefficients, ordered by increasing degree, including zeros. All modular residues use canonical representatives. For a positive divisor $D \mid P$, put

$$F_D(z) = \left\lfloor \frac{[z]_P}{D} \right\rceil. \tag{2}$$

This is a coefficient map, not division by a polynomial. The ring multiplication reduces degrees using $x^n = -1$; it is different from the ordinary middle-product windows of the NOISE construction.

Proposition 1 — Signed Toeplitz representation and cancellation. For polynomials a, b of length n, their integer negacyclic product has coefficients

$$(a \star b)_i = \sum_{j=0}^{i} a_j b_{i-j} - \sum_{j=i+1}^{n-1} a_j b_{n+i-j}. \tag{3}$$

Multiplication by a is represented by a signed negacyclic Toeplitz matrix. For $A \in R^{k\times k}$, $s, r \in R^k$,

$$(As)^T r = (A^T r)^T s. \tag{4}$$

Proof. A degree $i + n$ term of the ordinary product contributes with a minus sign to degree i, giving (3). Each matrix entry is a_{i-j} for $i \geq j$, and $-a_{n+i-j}$ otherwise. For (4), expand the left side as $\sum_{u,v} A_{uv} s_v r_u$ and commute scalar ring factors. The right side is the same double sum. This uses neither matrix symmetry nor invertibility. The integer identity reduces modulo P. □

The abstract public matrix has **nine independent uniform ring elements**, hence 2304 independent coefficients modulo P. Its expanded coefficient matrix has signed repeated entries within each block. Those repetitions do not turn it into the single overlapping Hankel ribbon of HB. A theorem for one public distribution is not a theorem for the other.

3. Exact laws and the base encryption scheme

Let $\chi = \mathrm{CBD}_1$, the difference of two independent fair bits:

$$\Pr[E = -1] = \Pr[E = 1] = \tfrac{1}{4}, \qquad \Pr[E = 0] = \tfrac{1}{2}. \tag{5}$$

Every displayed product of sampling laws means independent sampling of all its coefficients. Key-generation, encryption and lifting randomness are independent unless a dependency is explicitly specified.

KeyGen. Choose A uniformly in $R^{k\times k}$. Draw $s, e_0 \leftarrow \chi^{kn}$, and set

$$b = F_Q(As + e_0), \qquad pk = (A, b), \quad sk = s. \tag{6}$$

Here b is stored as kn integers in $[0, P/Q)$, not as an independently sampled noise vector.

Encrypt. For a message $\mu \in \{0,1\}^n$, draw $r, e_1 \leftarrow \chi^{kn}$, $e_2 \leftarrow \chi^n$, and $\rho \leftarrow \mathrm{Unif}(\{0, \dots, Q-1\}^{kn})$. Define

$$\begin{aligned} v &= Qb + \rho, \\ c_1 &= F_C(A^T r + e_1), \\ c_2 &= F_C(v^T r + e_2 + (P/2)\mu). \end{aligned} \tag{7}$$

The ciphertext is $c = (c_1, c_2)$. The lifted vector v is internal and is not transmitted. Both ciphertext components use the same r.

Decrypt. Validate lengths and coefficient ranges, compute

$$z = [Cc_2 - (Cc_1)^T s]_P, \tag{8}$$

and decode each coefficient by

$$\mathrm{DecBit}(z) = \mathbf{1}_{\{P/4 < [z]_P < 3P/4\}}. \tag{9}$$

Both distance ties are assigned zero. This defines a deterministic decryption result even when the ciphertext is dishonest. Invalid formats are rejected at the base-PKE interface. In the KEM interface they invoke a specified implicit-rejection branch.

In the actual key law v need not be uniform. The useful uniform-lifting identity belongs to an ideal-public-key game, not to real key generation by declaration. In particular, replacing v by Qb changes the encryption distribution and removes an ingredient of the later proof.

4. Exact residues and the decoding event

Let the actual coefficient remainders be

$$r_0 = [As + e_0]_Q, \quad r_1 = [A^T r + e_1]_C, \quad r_2 = [v^T r + e_2]_C. \tag{10}$$

Because $Q, C \mid P$, these are also the remainders after reduction modulo P. Since $C \mid P/2$, r_2 is independent of the message **at fixed coins**. This statement does not assert probabilistic independence of r_2 and e_2.

Lemma 2 — Signed error identity. For every honest choice of the algorithms' variables,

$$z \equiv (P/2)\mu + \varepsilon \pmod P, \quad \varepsilon = (e_0 + \rho - r_0)^T r + (r_1 - e_1)^T s + e_2 - r_2. \tag{11}$$

The right side can be evaluated using ordinary integer negacyclic products to obtain an integer lift of each error coefficient.

Proof. Euclidean division gives $Qb \equiv As + e_0 - r_0$, $Cc_1 \equiv A^T r + e_1 - r_1$, and $Cc_2 \equiv v^T r + e_2 + (P/2)\mu - r_2$. Substitute $v = Qb + \rho$ into (8). The two main products cancel by (4). The remaining terms give (11) with the stated signs. No random-variable independence is used in this identity. □

Lemma 3 — Strict decoding margin. If $z \equiv (P/2)\mu + e \pmod{P}$ for a bit μ and an integer e satisfying $|e| < P/4$, then $\mathrm{DecBit}(z) = \mu$. Thus every honest failure is contained in

$$\mathcal{F} = \{\max_i |\varepsilon_i| \geq P/4\}. \tag{12}$$

Proof. On the circle of circumference P, the centers 0 and $P/2$ are separated by $P/2$. The strict bound makes the correct center strictly nearer. Equivalently, its two possible neighborhoods are precisely those specified by (9). Apply this argument to each coefficient. The event $\mathcal{F}$ is a sufficient containing event, not an equivalent characterization of failure: a large integer lift can wrap back to a correctly decoded residue. □

Ordinary distance between canonical representatives is not the distance used here. Nor can one replace the true remainders by independent uniforms. For example, r_1 depends on both r and e_1, and r_0 depends on s, e_0, A. The proof below integrates in an order that retains these dependencies.

5. The correctness notion and conditioning order

For a generated key define its worst-message failure probability over fresh encryption coins. The quantity to be bounded is

$$\delta = \mathbb{E}_{pk,sk}\left[\max_{\mu\in\{0,1\}^n} \Pr[\mathrm{Dec}(sk, \mathrm{Enc}(pk, \mu)) \neq \mu \mid pk, sk]\right]. \tag{13}$$

The expectation includes the randomness that generated the key. This is the averaged maximum used by the external FO correctness definition [1]; it is not a maximum over every key.

The event (12) is the same for all messages when their encryptions are coupled with the same coins, because $C \mid P/2$. For every latent key-generation draw it contains all possible message failures. Taking the maximum over messages therefore occurs before the final average without multiplying by the number of messages. If several latent draws yield the same (pk, sk), condition on that key and average those latent draws; the containing-event argument still applies.

The proof works for any fixed matrix A, or any random matrix independent of s, e_0. Uniformity or nonsingularity of A is not a premise of correctness. Uniformity will be a separate premise of the source assumptions in the confidentiality proof.

6. Exponential moments without independent rounding errors

For $u > 1$, independent scalar variables $E, R, S \sim \chi$, and T uniform in $\{0, \dots, Q-1\}$, define

$$\begin{aligned} a(u) &= \mathbb{E}_{E,R,T} u^{(|E|+T)R}, \\ b(u) &= \mathbb{E}_S \max_{0\leq c<C} \mathbb{E}_E u^{S([c+E]_C - E)}, \\ c(u) &= \mathbb{E}_E u^E. \end{aligned} \tag{14}$$

The maximum is inside the expectation over S. Exchanging these operations would give a different quantity. The source letter c inside the maximum denotes an integer residue, not a ciphertext.

Lemma 4 — Pointwise domination of a hidden remainder. For fixed integer e, any $t \in [0, Q-1]$, and symmetric R independent of uniform T,

$$\mathbb{E}_{R,T} u^{(e+T-t)R} \leq \mathbb{E}_{R,T} u^{(|e|+T)R}. \tag{15}$$

The same bound holds when the exponent is negated.

Proof. The function $f(y) = \mathbb{E}_R u^{yR}$ is even and convex. After averaging over the uniform interval for T, the resulting function of $e + (Q-1)/2 - t$ is again even and convex: reflect T about its midpoint. Its maximum for $0 \le t \le Q-1$ is at an endpoint. The endpoint farthest from its symmetry center is $t = 0$ for $e \ge 0$, and $t = Q-1$ for $e \le 0$. Reflecting T in the latter case gives the right side. Symmetry of R absorbs a minus sign. This is pointwise in t; it remains valid when that remainder depends on other variables fixed during conditioning. □

Theorem 5 — Joint-error Chernoff bound. For the C1 laws and any matrix independent of key secrets and errors,

$$\mathbb{E}u^{\varepsilon_i} \le (a(u)b(u))^{kn}c(u), \qquad \mathbb{E}u^{-\varepsilon_i} \le u^{C-1}(a(u)b(u))^{kn}c(u). \tag{16}$$

Consequently,

$$\delta \le n(1+u^{C-1})(a(u)b(u))^{kn}c(u)u^{-P/4}. \tag{17}$$

Proof. Fix one output index. In each negacyclic inner product, the coefficient formula pairs every input coordinate with exactly one coordinate of the opposite factor, with a fixed sign $\sigma \in \{-1, 1\}$. Thus there are exactly kn pairs, not $(kn)^2$.

For the positive exponential, first use $-r_{2,i} \le 0$; for the negative exponential use $r_{2,i} \le C-1$. This introduces the factor 1 or u^{C-1} and allows $e_{2,i}$ to be averaged separately. By symmetry its moment in either direction is $c(u)$. No independence of r_2, e_2 is asserted.

Now fix A, s, e_0, r, ρ. For every coefficient of e_1, the relevant residue has the form $[c_j + e_{1,j}]_C$, with $c_j = [(A^T r)_j]_C$. The $e_{1,j}$'s are mutually independent and each appears once in the selected output. Integrate them first and bound each factor by $B_u(\sigma s_j) = \max_c \mathbb{E}_E u^{\sigma s_j([c+E]_C - E)}$. These upper bounds no longer depend on r. The signs may differ, but the indexing is a permutation of all coordinates of s.

With A, s, e_0 now fixed, r_0 is fixed. Every r_j and its paired ρ_j occur once in the first product, so independence of the *actual sampled* r, ρ, together with Lemma 4, bounds their moment by a product of factors depending only on the corresponding $|e_{0,j}|$. This removes all dependence on r_0 without assigning a law to it. Only then average over the independent coefficients of e_0 and s. The first product gives $a(u)^{kn}$; symmetry of S absorbs each sign in the second and gives $b(u)^{kn}$. The argument with all signs reversed is identical.

Markov's inequality bounds the upper and lower tails by the two quantities in (16) multiplied by $u^{-P/4}$. Sum them and use a union bound over n outputs. Output coordinates need not be independent. Section 5 justifies taking the maximum over messages inside the average. □

The complete proof uses finite expectations for C1. A replacement noise law would require recomputing all three functions and checking the same symmetry and independence premises.

7. A rational certificate and a genuine failure event

Corollary 6 — C1 failure certificate. At $u = 3/2$,

$$a = \frac{52261}{41472}, \qquad b = \frac{67}{64}, \qquad c = \frac{25}{24}, \tag{18}$$

and the C1 profile obeys

$$\delta \le 256\,\frac{5}{2}\left(\frac{52261}{41472}\frac{67}{64}\right)^{768}\frac{25}{24}\left(\frac{2}{3}\right)^{1024} < 2^{-282}. \tag{19}$$

Proof with a finite rational certificate. Substitute the three-point law (5) and the four possible values of T into (14). These finite rational sums give (18). If the positive rational right side of (19) is reduced to N/D, the accompanying two independently written checks verify the integer inequality $2^{282}N < D$. The logarithm approximately -282.661702381374 is only a display value, not the acceptance criterion. The general probability argument is Theorem 5; a sampled failure rate is not used. □

Counterexample 7 — Correctness is not perfect. For every matrix A, the C1 support contains an event of probability 2^{-6400} on which the all-zero message is decoded incorrectly. Therefore

$$2^{-6400} \le \delta < 2^{-282}. \tag{20}$$

Proof. Set every coefficient of s to 0, of e_0 to -1, and of r to 1; set e_1, e_2, ρ to zero. Then $r_0 = 3$, $Qb = 4092$, and the first lifted error coefficient is

$$\varepsilon_0 = -4 \cdot 3(1 - 255) = 3048 \equiv -1048 \pmod{4096}. \tag{21}$$

It lies strictly between 1024 and 3072 as a canonical residue, so it decodes to 1 although $\mu_0 = 0$. The event probabilities multiply to $2^{-768}2^{-1536}2^{-1536}2^{-768}2^{-256}2^{-1536} = 2^{-6400}$. The maximum in (13) is at least the failure probability for the zero message. Since the witness works for every A, no matrix-generation probability is charged. □

This event rules out setting the FO correctness error to zero. Its existence is not an efficient attack, and its extremely small probability does not justify ignoring it in a theorem that claims perfection.

Corollary 8 — Bad keys and reuse. If $d(pk, sk)$ is the conditional maximum in (13), then for every $\theta > 0$,

$$\Pr[d(pk, sk) > \theta] \le \min(1, 2^{-282}/\theta). \tag{22}$$

For at most r honest encryptions with fresh coins under a single generated key, including adaptively chosen messages, the probability of any decryption failure is at most $\min(1, r2^{-282})$.

Proof. The first claim is Markov's inequality applied to the nonnegative random variable d. Given a key, each fresh encryption has conditional failure probability at most d, regardless of the selected message. A conditional union bound gives rd, and averaging gives the second statement. No new key-generation randomness is attributed to each reuse. □

These are finite bounds for this fixed profile, not an asymptotic negligibility theorem for an unstated family. In particular, a bound on averaged keys is not a uniform bound for every adversarially selected key.

8. Two exact Module-LWE source problems

For $j \in \{3, 4\}$, define $\mathsf{DMLWE}^{\chi}_{3,j}$ by the two experiments

$$\begin{aligned} &\text{real:} \quad B \leftarrow \mathrm{Unif}(R^{j\times 3}), \quad t \leftarrow \chi^{3n}, \quad e \leftarrow \chi^{jn}, \quad (B, Bt + e), \\ &\text{ideal:} \quad B \leftarrow \mathrm{Unif}(R^{j\times 3}), \quad y \leftarrow \mathrm{Unif}(R^{j}), \quad (B, y). \end{aligned} \tag{23}$$

All initial draws are independent. The public matrix has independent ring elements; each coefficient of every secret and error is CBD1. Write $\epsilon_{3,j}(\mathcal{B})$ for the absolute difference of a distinguisher's acceptance probabilities. The secret is shared across the j module equations. The three- and four-row problems are separate source interfaces.

Neither a generic LWE matrix nor the HB source of the NOISE construction has this distribution.

The modulus is a power of two, and the exact small secret and error law is part of the assumption. No standard worst-case lattice theorem is declared to cover (23) merely because its name contains LWE.

Lemma 9 — Uniform lifting with the whole public view. If V is uniform in R^3 and independent of A, then the joint law of $(A, F_Q(V), V)$ equals that of

$$A, \quad b \leftarrow \mathrm{Unif}(\mathbb{Z}_{P/Q}^{3n}), \quad \rho \leftarrow \mathrm{Unif}(\mathbb{Z}_Q^{3n}), \quad V = Qb + \rho, \tag{24}$$

with the new draws independent. Conditional on (A, b), it remains a uniform lift after an adversary chooses messages and a state using independent coins and only that public view.

Proof. Coordinatewise quotient and remainder give a bijection from $\mathbb{Z}_P$ to $\mathbb{Z}_{P/Q} \times \mathbb{Z}_Q$. Apply it independently to all coefficients. Independence of A, V makes the remainder independent of (A, b), not only of b. Further conditioning on a randomized function of this public view and independent adversary coins preserves the conditional remainder law on every positive-probability fiber. □

The independence qualification is essential. Independence from the quotient alone would not imply that the hidden lift remainder stays uniform after conditioning on the complete public view. The simulation below satisfies the stated stronger premise because the matrix rows and the retained uniform lift are sampled independently.

9. Conditional CPA through two joint simulations

Use $\mathrm{Adv}_{CPA} = |\Pr[\hat{\beta} = \beta] - 1/2|$. The adversary receives an honest public key, chooses two messages and then receives one challenge. Public encryption queries can be generated locally; there is no decryption oracle.

Theorem 10 — Abstract full-matrix CPA reduction. For every classical adversary $\mathcal{A}$, two distinguishers at its augmented simulation budget satisfy

$$\mathrm{Adv}_{CPA}(\mathcal{A}) \le \epsilon_{3,3}(\mathcal{B}_1) + \epsilon_{3,4}(\mathcal{B}_2). \tag{25}$$

Proof. In game G_0, generate the real key and ciphertext. In G_1, keep a uniform matrix A but make b independently uniform in the quotient space; encrypt with the unchanged algorithm. A distinguisher given (A, y) from (23) with three rows publishes $(A, F_Q(y))$, runs the adversary, selects the challenge bit and publicly encrypts. It returns the success indicator. Real and ideal source branches reproduce G_0, G_1 exactly. It does not need the source secret.

For the second transition receive (B, y) with four rows. Write its first three rows as A^T and its last row as V^T. Publish $pk = (A, F_Q(V))$, retaining V. The matrix A and the uniform V are independent. Once the adversary chooses messages, select β and return

$$c_1 = F_C(y_{0:3}), \qquad c_2 = F_C(y_3 + (P/2)\mu_\beta). \tag{26}$$

In the real source, all four responses use the same secret t; take it as the encryption secret r. The first three errors give e_1, and the fourth gives e_2, with their required independence. Lemma 9 gives the full conditional law of $V = Qb + \rho$, even after message selection. Thus the output is precisely the challenge distribution in G_1. Other local encryptions use their own fresh lifts, not the retained V.

In the ideal source, the four outputs are jointly uniform and independent of the complete matrix. Translation by the message preserves uniformity, and equal-width floor cells preserve uniform quotients. This is game G_2, in which the challenge is independent of β even given the key and

chosen messages. Its success probability is 1/2. The two absolute differences of game-success probabilities are exactly the two source advantages, proving (25) by the triangle inequality. □

This is a reduction, not a proof that either source advantage is small. It uses one challenge of each stated kind in two different reductions, not an oracle supplying unlimited equations under a hidden secret. Ideal games need no decryption secret. Correctness and secrecy are established by different arguments.

10. Seeded matrix generation and programming

The seeded representation uses a 256-bit seed and an independent oracle $\mathcal{A}$. Each entry is obtained at an injective input identifying the seed and its row and column. A response contains 512 bytes, interpreted as 256 little-endian 16-bit words. Keep each word modulo 4096. Distinct matrix-entry inputs have independent responses.

Proposition 11 — Exact matrix expansion in the ideal-oracle model. This procedure produces the required independent uniform matrix for a fresh seed. A simulator that is given a uniform matrix can program it at a fresh seed with an exactly matching raw-byte distribution. With at most q_A^{pre} prior matrix-oracle queries, the conflict probability is at most

$$\epsilon_A = \min(1, q_A^{pre} 2^{-256}). \tag{27}$$

The seeded CPA reduction consequently has the conservative bound

$$\mathrm{Adv}_{CPA}^{seed} \le \epsilon_{3,3} + \epsilon_{3,4} + 2\epsilon_A. \tag{28}$$

Proof. Every 12-bit value has exactly 16 preimages among 16-bit words. To program a supplied coefficient, append an independent uniform four-bit high part; this samples its full conditional raw-word fiber. Repeat independently across all coefficients and entries. A prior query can involve at most one seed at the prescribed domain. The number of previously exposed distinct seeds is therefore at most the query count. A fresh uniform seed hits that set with probability at most (27). Abort on a conflict rather than resampling until success; otherwise the seed law would be silently conditioned. Outside the bad event, the programmed oracle and the public quotient have the required joint law. Apply this construction separately in the two reductions of Theorem 10 and add the two allowances. □

The raw 512-byte responses matter: programming only their low bits without sampling the discarded bits would not simulate an adversary who queries the entire oracle response. Unused matrix-oracle inputs are answered independently and consistently. This proof is not a claim that a short public seed and fixed SHAKE256 output are statistically identical to an explicitly sampled long matrix. Concrete hash instantiation is a separate modeling obligation.

11. Random tapes, deterministic encryption and implicit rejection

The C1 specification uses a 2176-byte encryption tape. The first 1792 bytes produce the $7n$ CBD1 coefficients of r, e_1, e_2: use the difference of the two low bits of each byte. The last 384 bytes produce 768 nibbles, each reduced modulo four, for ρ.

Proposition 12 — Exact coin law. Uniform bytes under this allocation give precisely the independent distributions in (7).

Proof. Two independent fair bits have difference law (5). Different bytes use independent bit pairs. The two nibbles of one uniform byte are independent uniform four-bit values; reducing each modulo four leaves independent uniform two-bit values. All remainder bytes are disjoint

from the CBD bytes. The counts are $3n + 3n + n = 1792$ CBD coefficients and $3n = 768$ remainders. Ignoring unused bits changes neither law. □

Use independent classical random oracles G, H, separate also from $\mathcal{A}$. Let $G(pk, \mu)$ give the tape and $H(pk, \mu, c)$ give a 256-bit key. Set

$$\mathrm{Enc}_1(pk, \mu) = \mathrm{Enc}(pk, \mu; G(pk, \mu)). \tag{29}$$

KEM key generation additionally draws one fallback secret z_0 uniform on $\{0,1\}^{256}$, independently. Encapsulation samples uniform μ, returns $c = \mathrm{Enc}_1(pk, \mu)$ and $K = H(pk, \mu, c)$. Decapsulation decrypts to μ', re-encrypts with $G(pk, \mu')$, and returns

$$\begin{cases} H(pk, \mu', c), & \text{canonical parsing succeeds and } \mathrm{Enc}_1(pk, \mu') = c, \\ H(pk, z_0, c), & \text{otherwise.} \end{cases} \tag{30}$$

The equality is byte-exact. No accepted alias is normalized into the challenge encoding. Wrong lengths take the fallback branch. The fallback is drawn once per key, not afresh per rejection. These rules preserve repeat-query consistency. Hashing the actual ciphertext inside H is part of this chosen transform.

At the abstract level the oracle names provide domain separation; raw strings must implement injective encodings of the tuples. Public-key and message lengths are fixed and the ciphertext is the final field. The accompanying programs verify base-PKE arithmetic, coefficient encodings and random-tape distributions. They do not implement the random oracles G and H or their tuple encodings. Injective encodings and domain separation remain explicit requirements for a future KEM implementation; the security theorem here concerns the abstract independent-oracle model.

12. CCA composition with a nonzero correctness term

The external ingredient is the corrected classical-ROM analysis of HHK [1], using transformation T followed by implicit-rejection $U^{\not\perp}$, with the ciphertext included in H. The corrected Theorem 3.2 counts plaintext-checking queries in its correctness term. With no ciphertext-validity oracle, the OW-PCA route does not require spreadness. Theorem 3.4 then gives the KEM step. These are imported published results, not newly invented transforms.

Let q_G, q_H count explicit adversarial queries and q_D count decapsulations. For an implementation that evaluates both candidate hashes, conservative totals are

$$Q_G = q_G + q_D + 1, \quad Q_H = q_H + 2q_D + 1, \quad Q_{RO} = Q_G + Q_H = q_G + q_H + 3q_D + 2. \tag{31}$$

The additions cover internal calls and the honest challenge. Counting repeated inputs only increases this upper bound. An invalid length needs at most the fallback hash, which is already covered. Oracle input length and arithmetic work also count toward the reduction's resources.

Theorem 13 — Conditional single-user CCA bound. In the ideal classical oracles, for the challenge-first KEM experiment of the cited composition, appropriate source distinguishers satisfy

$$\begin{aligned} \mathrm{Adv}_{CCA} \le\ & 3\epsilon_{3,3}(\mathcal{B}_1) + 3\epsilon_{3,4}(\mathcal{B}_2) + 6\epsilon_A \\ & + Q_{RO}2^{-282} + 3Q_{RO}2^{-256}. \end{aligned} \tag{32}$$

The KEM advantage is success bias from 1/2; source advantages are acceptance gaps.

Proof by application of the identified external theorem. The base PKE has exactly the averaged

worst-message correctness notion (13), with $\delta < 2^{-282}$. The random tape has its required uniform law by Proposition 12. Re-encryption implements T, and a single independent fallback with (30) implements the chosen implicit-rejection U. For a fixed single-user public key, prefixing pk restricts the independent random functions to an injectively encoded domain.

The corrected HHK composition bounds the bias by $Q_{RO}\delta + 3Q_{RO}/|\mathcal{M}| + 3\,\mathrm{Adv}_{CPA}$, where $|\mathcal{M}| = 2^{256}$. Its proof covers the faulty-decryption events; none are assumed absent here. The interface uses plaintext checking, not an unprovided validity oracle. Substitute (28), use (19), and retain the total query budgets (31). All source distinguishers are evaluated at the actual augmented simulation time and query budget. This proves the displayed specialization of the imported theorem. □

An explicit-call shorthand is insufficient unless implicit calls are included in the oracle budget. Likewise, writing $\delta = 0$ would contradict Counterexample 7. No QROM theorem, multi-user guarantee, implementation leakage guarantee or concrete SHAKE theorem follows from (32).

Proposition 14 — An allowance for decapsulation before the challenge. To extend the challenge-first experiment to a phase with at most d_{pre} decapsulation queries before a fresh uniform encapsulation message is selected, a sufficient additional allowance is

$$\epsilon_{phase} \le \min(1, d_{pre}2^{-256} + \delta). \tag{33}$$

Proof. Presample the challenge with its message hidden during the preliminary phase, using the same fixed oracle functions. Its disclosure may be delayed because the phase receives no challenge information. Abort only if a preliminary query equals the hidden challenge ciphertext. For fixed keys and oracles, any ciphertext has at most one **successfully decrypting** message preimage under Enc_1: deterministic decryption cannot equal two different messages. Thus the union of d_{pre} queried ciphertexts accounts for at most d_{pre} good messages. All remaining collisions require a message whose deterministic encryption decrypts incorrectly. The expected fraction of such messages, over keys and independent G, is at most δ: for each message its oracle tape is uniform and its average failure is bounded by (13). A fresh uniform message is independent of the preliminary phase when the functions and key are fixed. Averaging gives (33), including adaptive preliminary queries. Outside this event the challenger can relay the phase exactly. □

This extra allowance is stated separately from the imported main-game bound. Perfect correctness, which could remove its final δ, is unavailable. It is also separate from matrix-seed prequeries. The argument is about an honestly generated single key, not arbitrary related-key or malicious-key interfaces.

13. Encodings, storage, and the meaning of “under one kilobyte”

Proposition 15 — C1 payload sizes. With a one-byte profile identifier and a 32-byte matrix seed, the public key has 993 bytes; the ciphertext has 1408 bytes:

$$\begin{aligned} |pk| &= 1 + 32 + 3 \cdot 256 \cdot 10/8 = 993, \\ |c| &= (3+1) \cdot 256 \cdot 11/8 = 1408. \end{aligned} \tag{34}$$

Proof. Each public quotient is in $[0, 4096/4)$, requiring ten bits. Each ciphertext quotient is in $[0, 4096/2)$, requiring eleven bits. There are 768 public quotients and 1024 ciphertext coefficients. Concatenate fixed-width fields coefficient-first and little-endian within the bit stream. The two totals are byte-aligned, so there are no padding aliases. The seed and identifier contribute 33 bytes only to the public key. □

The mathematical full-matrix representation would require 3456 bytes for the nine polynomials alone, plus 960 bytes for b, or 4416 bytes before metadata. The reduction to 993 bytes uses the seeded generation model, not a lossless compression of every arbitrary independent full matrix into 32 bytes. A KEM secret key must retain enough public data to re-encrypt, as well as the fallback and secret vector; its complete byte format is not specified by the two public payload numbers.

The local test encoder does not claim to reproduce the historical profile identifier or literal oracle-domain strings. Both need checking against the original byte specification before a release. Generic parsers in the tests reject wrong lengths and nonzero padding on auxiliary profiles; C1's ciphertext itself has no unused bits. Parameters and profile negotiation remain outside the payload count except for the declared identifier.

A raw PKE ciphertext remains malleable: translating one $c_{2,i}$ by $P/(2C)$ adds $P/2$ to the corresponding decryption residue. Away from ties this complements its decoded bit. In the KEM, such a valid-format change is subjected to the re-encryption check; merely remaining a canonical ciphertext is not acceptance. The FO proof, not a format check or a handful of rejected modifications, supplies the conditional CCA claim.

14. Concrete resources are additional hypotheses

Suppose, only for illustration, that both exact source problems satisfy a quantitative envelope

$$\epsilon_{3,j}(\mathcal{B}) \le T_{\mathcal{B}} 2^{-s_*}, \qquad T_{\mathcal{B}} \le \kappa_* T, \tag{35}$$

in one explicitly chosen work unit. Let $T \ge 1$, $q_G + q_H + q_D \le T$, and $q_A^{pre} \le T$. Neither this envelope nor a concrete value of κ_* is proved by a lattice-cost estimate.

Proposition 16 — Conditional work/advantage calculation. For the main game of Theorem 13 and normalized advantage $a_{CCA} = 2\,\mathrm{Adv}_{CCA}$,

$$\frac{a_{CCA}}{T} \le 12\kappa_* 2^{-s_*} + 10 \cdot 2^{-282} + 42 \cdot 2^{-256}. \tag{36}$$

Proof. Since $Q_{RO} \le 3T + 2 \le 5T$, substitute (35) into (32). The two source terms give $6\kappa_* T 2^{-s_*}$ before normalization. The matrix-programming term contributes at most $6T2^{-256}$, and the hash-guessing term at most $15T2^{-256}$. Multiply the complete inequality by two and divide by T. The correctness term becomes at most $10 \cdot 2^{-282}$. □

Assumed $\log_2 \kappa_*$	0	10	16	20	24
Required integer s_*	132	142	148	152	156

The table gives the smallest integer s_* making (36) strictly smaller than 2^{-128}. It is verified by exact rational comparisons, including failure at the preceding integer. It is conditional arithmetic, not a security certificate. A cost for one known attack at a particular success probability is not an upper bound on the advantage of **every** adversary. Source attack work, bit operations, preprocessing, oracle traffic and Python time are not interchangeable. Applying Proposition 14 adds its phase allowance; the table does not silently include it.

Earlier exploratory estimator runs are not used as evidence here. Their models need not cover every module-aware attack, and no numerical estimator output is promoted to a security certificate. Current primary metadata for structured cryptanalysis [5] supports the need to preserve module symmetries, but does not evaluate the exact C1 scheme or certify any concrete level.

15. Exact CBD1 guessing and a fixed-character calculation

Proposition 17 — Optimal prior guessing in the CBD1 law. For a vector of ζ independent CBD1 coefficients, every particular vector of nonzero weight w has probability $2^{-(\zeta+w)}$, and there are $\binom{\zeta}{w}2^w$ such vectors. Consequently,

$$\Pr[W \le w] = 2^{-\zeta}\sum_{j=0}^{w}\binom{\zeta}{j}, \qquad N_{\le w} = \sum_{j=0}^{w}\binom{\zeta}{j}2^j. \tag{37}$$

To reach target prior probability θ, order weights increasingly. If θ enters shell w, the smallest number of guesses is

$$N_{<w} + \left\lceil (\theta - \Pr[W < w])2^{\zeta+w} \right\rceil. \tag{38}$$

Proof. Zero has mass $1/2$, each sign has mass $1/4$. Multiply coefficient probabilities and count the supports and signs. All vectors in a lighter shell have strictly greater individual probability. Exchanging a lower-probability guess for an unguessed higher-probability vector cannot decrease the covered mass, so a prefix of this order is optimal. The last shell has constant individual mass, giving the ceiling. □

This is a prior-distribution calculation with no public-key information. It is not the cost of an optimal public-key attack. In particular, the exact Shannon entropy 1.5ζ and min-entropy ζ do not replace the cost/success curve or account for conditioning on a public observation.

For a fixed integer frequency h, direct averaging gives

$$\mathbb{E}_{E\sim\chi}\exp(2\pi i h E/P) = \cos^2(\pi h/P). \tag{39}$$

For a centered representative $|h| \le P/2$, this is at most $\exp(-\pi^2 h^2/P^2)$. Indeed, $\log\cos x \le -x^2/2$ for $|x| < \pi/2$, as follows by differentiating and using $\tan x \ge x$ for $x \ge 0$; the endpoints give zero directly. This controls one fixed character. It does not price the search for a useful dual vector, a decoding step or an adaptive family of tests.

16. Module symmetries and overlapping cyclic events

Multiplication by x^t in R is a signed permutation of coefficients. Thus the marginal CBD1 laws are preserved, and

$$x^t(Bs + e) = B(x^t s) + x^t e. \tag{40}$$

All these observations come from one instance; they are not new independent source samples. For sign-insensitive events there are n, not kn, simultaneous cyclic positions supplied by this operation. Independent rotation of module components would require a different transformation of the public matrix and its analysis.

If J events each have marginal probability p, then $p \le \Pr[\text{at least one}] \le \min(1, Jp)$. No formula $1-(1-p)^J$ follows without independence, and no universal $\log_2 J$ attack saving follows from the union bound. Shared preprocessing and dependence must be part of a specific attack analysis.

Proposition 18 — Exact cyclic zero-run probability. For a length-N CBD1 polynomial and $1 \le \zeta \le N$, define the integer transition matrix T_ζ on states $0, \dots, \zeta-1$: every state has one transition to 0 for a nonzero coefficient, and state j has one transition to $j+1$ for a zero if $j+1 < \zeta$. Then

$$\Pr[\text{some cyclic run of } \zeta \text{ zeros}] = 1 - 2^{-N}\operatorname{tr}(T_\zeta^N). \tag{41}$$

Proof. The zero/nonzero indicators are independent fair bits. For an indexed cyclic word

containing a nonzero symbol and no forbidden run, the state before each symbol is uniquely determined by its number of immediately preceding zeros. It gives exactly one compatible closed walk at that indexed starting position. Conversely every counted walk gives such a word. The all-zero word is forbidden because $\zeta \leq N$. The trace counts all permitted indexed words; divide by 2^N and take the complement. □

For $N = 256, \zeta = 16$, the exact rational probability is approximately 0.0019514643. The independent-window expression is almost twice as large. Both are computed for this explicitly defined event; no claim is made that it models every primal or dual hybrid strategy. The accompanying two implementations use matrix exponentiation and dynamic closed-walk counting, respectively, and agree with exhaustive enumeration on small domains.

17. Ramified projections and explicit-noise boundaries

Proposition 19 — Low ramified quotients carry uniform CBD1 error. In the C1 ring, put $w = x + 1$. Then $(w^{256}) = (2)$ and $w^{3072} = 0$. For $1 \leq t \leq 256$,

$$R/(w^t) \simeq \mathbb{F}_2[w]/(w^t), \tag{42}$$

and the image of an independently sampled CBD1 polynomial is uniform on this quotient. For an unrounded source in (23), its projected noisy responses are consequently uniform independently of the secret, conditional on the public matrix.

Proof. Expand $(w-1)^{256} + 1$. Every intermediate binomial coefficient is even, so the defining relation gives $w^{256} = -2g(w)$, where $g(0) = 1$. First $w^{256} \in 2R$ implies $w^{3072} = 0$. Thus w is nilpotent, and $g(w) = 1 + wh(w)$ is a unit by a finite geometric series. This proves equality of the ideals. For $t \leq 256$, $(w^{256}) \subseteq (w^t)$, so 2 vanishes in the quotient. The resulting presentation is exactly (42).

The parity of each CBD1 coefficient is a fair bit. The change of polynomial coordinates from x to $w - 1$, modulo two, is an invertible triangular transformation on 256 coefficients. A uniform vector remains uniform; projecting to its first t coordinates is uniform on $\mathbb{F}_2^t$. Independent error polynomials remain independent. Adding any fixed projected Bs preserves this uniformity. □

This proves a property of these **unrounded low quotients**, not secrecy of the entire source. Floor division is not a ring homomorphism. For instance, the integers 0 and 4 have the same parity but distinct quotients after division by 4. High-bit or carry-based attacks are not excluded by Proposition 19.

Proposition 20 — Explicit noise cannot simply be deleted. For $2 \leq Q < P$, $Q \mid P$, a fixed integer z and $E \sim \mathrm{CBD}_1$,

$$\Pr[F_Q(z + E) \neq F_Q(z)] = \tfrac{1}{4}\left(\mathbf{1}_{[z]_Q=0} + \mathbf{1}_{[z]_Q=Q-1}\right). \tag{43}$$

If $[z]_Q$ is uniform, its mean is $1/(2Q)$.

Proof. The zero error preserves the cell. A displacement -1 changes the quotient exactly at a cell's first point; a displacement $+1$ does so exactly at its last point. Since there are at least two cells, a wrap at 0 or $P - 1$ changes the output as well. Each nonzero error has probability 1/4. Averaging over the Q possible phases proves the mean. □

At $Q = 4$ that mean is 1/8, but no uniform-phase claim about As is used. A low-noise scheme can still be amenable to Boolean methods; a distribution mismatch with a noiseless analysis is not a defense theorem. This article supplies no new attack-cost estimate or source-hardness proof.

18. Reproducibility, evidence limits, and synthesis

The package contains a **new mathematical reference**, `a6_blindage.py`, and two new verifiers. The primary implementation uses direct signed negacyclic multiplication with arbitrary-precision integers. The independent verifier imports neither it nor the main verifier: it reconstructs modular products by carry-separated packed-integer convolution, signed errors by a separately written full convolution, and ciphertext encoding by an explicit bit stream. Its finite probability calculations use individual Bernoulli atoms rather than the primary distribution dictionary.

Verification family	Extent executed
Module double-product cancellation	200 cases
Strict decoder inputs	238
Signed error and all-message checks on small fixtures	6,672 encryptions
Pointwise hidden-remainder moment inequalities	1,232
C1-dimensional functional encryptions	12
Uniform-lift atoms	328
Complete auxiliary second-hybrid laws	64 laws; 3,456 weighted states
Public-quotient encoding round trips	100
Auxiliary canonical byte containers	256
Decoder complement residues at $P = 4096$	4,096
Cyclic run comparisons	55 profiles; 18,434 words across tests
Exact prior-guess quantiles versus full enumeration	24
Low-quotient uniformity	14 profiles; 2,120 bit vectors
Scalar explicit-noise boundary cases	240

The independent verifier additionally reconstructs one complete C1-dimensional synthetic transcript, its 256 signed error coordinates and 1408 encoded ciphertext bytes. It checks the rational certificate independently and fully enumerates 81,920 scalar latent states for whole-joint moment bounds on 24 fixed-public profiles. Those scalar experiments are auxiliary checks, not C1 security profiles. It also rechecks 64 second-hybrid laws and 45 cyclic profiles; overlapping domains are not counted as new independent cryptographic instances.

The certificate, rather than the twelve functional encryptions, proves the tiny failure bound. All transcripts intentionally expose their synthetic secrets and randomness. The scripts do not implement a production KEM, verify adaptive CCA security by testing, establish a quantum guarantee, or run lattice reduction at security-scale dimensions. No numerical estimator output is regenerated in this article.

The public package contains the article, public-source notes, the mathematical reference implementation, two independent verification paths, deterministic result files, a synthetic transcript, build scripts, and integrity hashes.

The JSON files omit timing and environment-dependent version fields. Mathematical acceptance uses exact integer or rational comparisons. The finite checks support the stated identities, formats, certificate arithmetic, and small-domain distribution claims; they are not a substitute for the two Module-LWE hardness assumptions or for adaptive CCA proofs.

Run from the package directory:

```
python3 verify_a6.py --output A6_RESULTS_RECHECK.json
python3 verify_a6_independent.py --output A6_INDEPENDENT_RECHECK.json
bash build_and_verify.sh
```

Scientific conclusion. C1-P12-CBD1 has an explicit nonzero correctness bound and a conditional classical-ROM route to a CCA KEM. It does not have an unconditional confidentiality theorem, a concrete 128-bit certificate, a QROM theorem, or a deployment recommendation. Its seeded public representation is exact only in the stated oracle model. The later CBD2 branch is outside the scope of this article and receives none of C1's certificate or source assumptions by inheritance.

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